

Maxwell Patterson

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SUMMARY

Data Science professional with experience in both client facing and non-client facing roles. Educational background includes an M.S. in Data Science from The University of Texas at Austin, as well as a B.S. in Mathematics and Physics from Dickinson College. Earned a professional certificate in Data Science from Northwestern University and have 8+ years of professional experience in data science and data analysis. Experienced in Python, R, SQL, and Tableau to create full-stack data solutions.

EDUCATION

UNIVERSITY OF TEXAS- AUSTIN

Master of Science, Data Science

May 2024

- **Coursework:** Machine Learning, Probability & Statistics, Data Structures & Algorithms, Causal Inference, Regressions, Advanced Predictive Models, Natural Language Processing, Deep Learning, Data Visualization, Reinforcement Learning

NORTHWESTERN UNIVERSITY

Professional Certificate, Data Science & Data Visualization

February 2020

DICKINSON COLLEGE

Bachelor of Science, Mathematics and Physics

May 2016

TECHNICAL SKILLS

- **Programming:** Python (Pytorch, NumPy, Pandas, Requests, Flask, Scikit-Learn, Matplotlib, PySpark), SQL (Databricks SQL, PostgreSQL, Microsoft SQL Server), R (ggplot, tidyverse, tidymodels, shiny), JavaScript (Plotly, D3, Leaflet)
- **Machine Learning:** Deep Learning Networks, Random Forests, Principal Component Analysis, K-Nearest Neighbors, K-Means, Linear/Logistic Regressions, XGBoost
- **Tools:** Databricks, Posit/RStudio, Alteryx, Tableau, Looker, Jupyter Notebooks, HTML5, CSS3, Git, SQL Server Management Studio, Amazon Redshift

PROFESSIONAL EXPERIENCE

SUFFOLK CONSTRUCTION

Senior Data Scientist

Roxbury, MA

March 2022 – Present

Data Scientist

Senior Data Analyst

- Designed and implemented a Shiny application with an LLM-powered chat interface to generate construction project schedules, integrating agent-based task generation with machine learning models for task duration estimation
- Trained and productionized a classification model that identifies construction projects likely to have a safety incident, which increased accuracy from prior model versions by 17%. Additionally created a Shiny app that displays model information for stakeholders. The fiscal quarter after its initial implementation was the best quarter for incident rates in company history
- Led analysis of historical project data to evaluate staffing estimation accuracy, revealing systemic underestimations that drove ~\$25M in annual lost revenue and informing changes to workforce planning processes

ZIPCAR

Boston, MA

Data Analyst

September 2021 – March 2022

- Uncovered ~\$70 in missing revenue per user for a subset of members by diagnosing issues in a dbt-based LTV estimation model, improving financial accuracy and reliability of revenue reporting
- Analyzed search behavior in key markets to identify previously active Zipcar members who disengaged, providing actionable insights to product and marketing teams for retention strategies
- Developed a Looker dashboard for the billing team to track KPIs on billing accuracy and success, enabling data-driven improvements in operational performance

NIELSEN IQ

Chicago, IL

Analytical Framework Consultant

August 2019 – September 2021

- Led project that used Python and SQL to develop a reporting tool hosted on a FastAPI endpoint. Using product information from both database and API data sources, this reporting tool had built-in data validation functionality based on defined use cases, and it reduced manual analysis time by 95%
- Researched NielsenIQ product data to create database prototypes based on the needs of retailers and manufacturers in the CPG industry. Met with client SMEs to confirm databases meet contractual requirements
- Member of an efficiency team within the organization that aimed to identify manual processes that could be replaced with Python scripts to reduce the length of product delivery timelines

TRILOGY EDUCATION SERVICES

Chicago, IL

Teaching Assistant, Data Visualization Program

April 2020 – August 2020

- Assisted students in a 24-week boot camp that teaches them Python, SQL, noSQL, JavaScript, Tableau, and machine learning concepts
- Reviewed and graded code submitted by students. Provided feedback in a timely manner

REVENUE SOLUTIONS, INC.

Chicago, IL

IT Consultant

September 2016 – August 2019

Associate IT Consultant

- Recognized with a company award for excellence in client communication and relationship management
- Provided client-site production support for SQL Server-based web applications that streamlined tax processing for state, city, and county government agencies, ensuring high availability and client satisfaction
- Led recurring client meetings to present progress on product enhancements, gather feedback, and align technical solutions with business needs
- Authored software requirements specifications (SRS) for custom ETL interfaces involving data from multiple tax types, each with unique business rules, enabling clients to integrate external data sources and improve accuracy in tax processing